Economic Applications of Game Theory

Chapter 2: Decision Theory

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The first lecture treated payoffs as given. This chapter explains what those payoffs mean. A game-theoretic payoff is not merely a ranking of final outcomes; it is a compact representation of how a player compares uncertain consequences induced by beliefs about the other players' actions.

2.1 Choice under certainty

Start with a set X of mutually exclusive, exhaustive alternatives. A decision maker's weak preference relation $\succeq$ compares alternatives: $x \succeq y$ means that x is at least as good as y.

> **Definition.** Preference relation. A relation $\succeq$ on X is a preference relation if it is complete and transitive. Completeness says that for every $x, y \in X$, either $x \succeq y$ or $y \succeq x$. Transitivity says that $x \succeq y$ and $y \succeq z$ imply $x \succeq z$.

Strict preference and indifference are derived from weak preference:

> **Definition.** For alternatives $x, y \in X$,
>
> $$x \succ y \iff [x \succeq y \text{ and not } y \succeq x], \qquad x \sim y \iff [x \succeq y \text{ and } y \succeq x].$$

A utility function assigns numbers to alternatives so that better alternatives receive larger numbers.

> **Theorem 2.1.** If X is finite, a relation on X can be represented by a utility function $u : X \to \mathbb{R}$,
>
> $$x \succeq y \iff u(x) \geq u(y),$$

if and only if the relation is complete and transitive. If u represents $\succeq$ and $f : \mathbb{R} \to \mathbb{R}$ is strictly increasing, then $f \circ u$ represents the same preferences.

The last sentence is the key. Utility under certainty is ORDINAL: only the order matters. The numerical gaps between utility values carry no independent meaning.

2.2 Decision making under uncertainty

Let Z be a finite set of prizes. A LOTTERY is a probability distribution $p : Z \to [0, 1]$ with $\sum_{z \in Z} p(z) = 1$. For example, the lottery below gives 10 with probability $1/2$ and 0 with probability $1/2$.

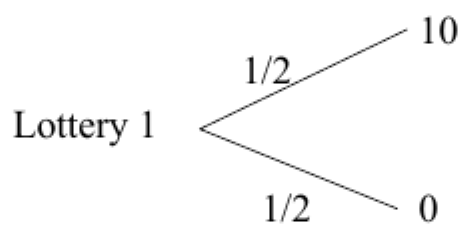


Figure 2.1:

In games, players usually do not know what the others will do. If a player has beliefs about the others' actions, each of his own actions induces a lottery over final outcomes. This is why game theory needs a theory of preference over lotteries, not just over sure prizes.

DEFINITION. VON NEUMANN-MORGENSTERN UTILITY. A preference relation $\succeq$ over lotteries P is represented by a VNM utility function $u : Z \to \mathbb{R}$ if, for every $p, q \in P$,

$$p \succeq q \iff U(p) \equiv \sum_{z \in Z} u(z)p(z) \geq \sum_{z \in Z} u(z)q(z) \equiv U(q).$$

This definition has two layers. First, U is an ordinary utility representation of preferences over lotteries. Second, it has the special expected-utility form: the utility of a lottery is the probability-weighted average of utility over prizes. For the lottery above, $U = \frac{1}{2}u(10) + \frac{1}{2}u(0)$.

The three axioms

The VNM representation is not automatic. It follows from three assumptions about preferences over lotteries.

> **Axiom 2.1.** Preferences over lotteries are complete and transitive.

> **Axiom 2.2.** Independence. For any lotteries p, q, r and any $\alpha \in (0, 1]$,
>
> $$\alpha p + (1-\alpha)r \succ \alpha q + (1-\alpha)r \iff p \succ q.$$

Independence says that mixing both options with the same background lottery should not reverse the comparison. One interpretation is dynamic consistency: if the decision maker prefers p to q, he should still prefer "first toss a coin, then get p on heads and r on tails" to the corresponding compound lottery with q replacing p.

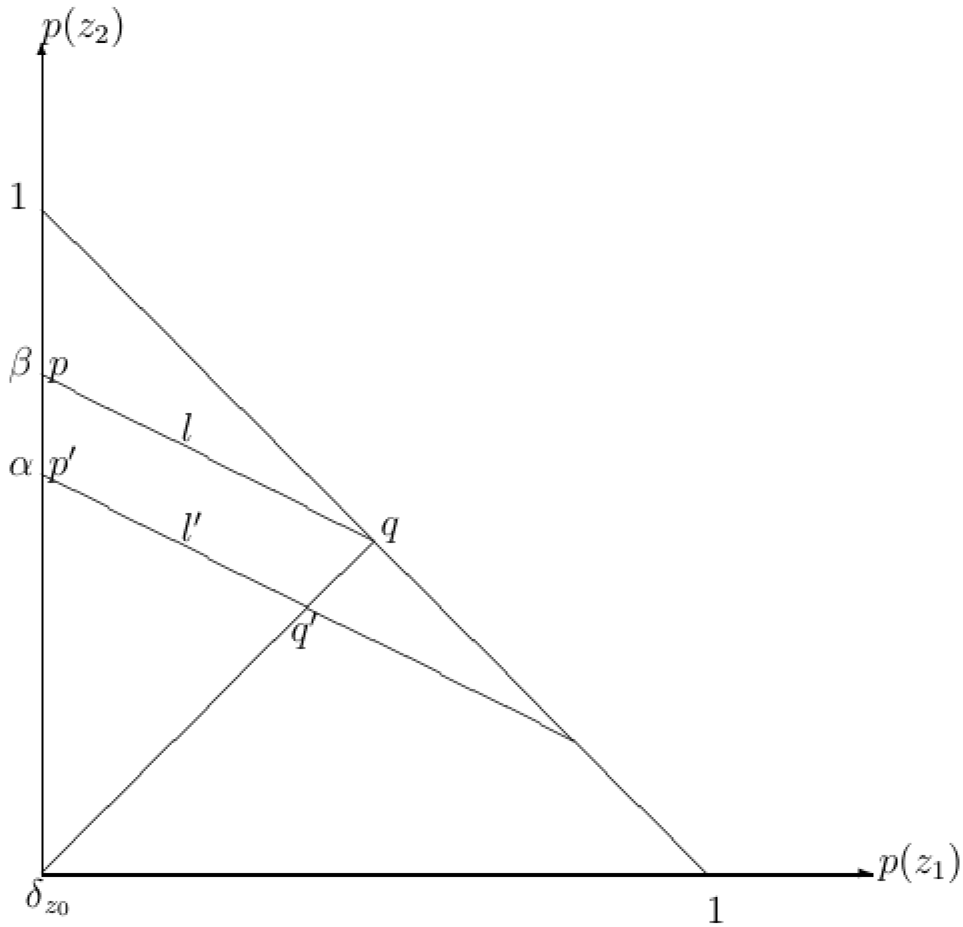
> **Axiom 2.3.** Continuity. If $p \succ q$, then for any third lottery r there are $\alpha, \beta \in (0, 1)$ such that
>
> $$\alpha p + (1-\alpha)r \succ q \quad \text{and} \quad p \succ \beta q + (1-\beta)r.$$

Continuity rules out infinitely good or infinitely bad prizes. It is mostly technical, but it makes the utility representation finite and well behaved.

Independence has a geometric implication. If $p \sim q$, then for any r and $\alpha \in [0, 1]$,

$$\alpha p + (1-\alpha)r \sim \alpha q + (1-\alpha)r.$$

With three prizes, lotteries can be drawn as points in a triangle. The implication above makes indifference curves straight, and repeating the argument from different base lotteries makes them parallel.



INDIFFERENCE CURVES OVER LOTTERIES ARE STRAIGHT AND PARALLEL UNDER INDEPENDENCE.

THEOREM 2.2. A relation $\succeq$ on the set of lotteries P has a VNM representation if and only if it satisfies Axioms 2.1-2.3. Moreover, two VNM utility functions u and $\tilde{u}$ represent the same preferences if and only if $\tilde{u} = \alpha u + \beta$ for some $\alpha > 0$ and $\beta \in \mathbb{R}$.

VNM utility is therefore CARDINAL in a limited sense. Positive affine transformations preserve preferences over lotteries; arbitrary increasing transformations need not. This is the difference between saying "only the order of sure prizes matters" and saying "utility gaps matter when computing expected utility."

2.3 Modeling strategic situations

Consider two players, Alice and Bob, with strategy sets S_A and S_B. If Alice plays s_A and Bob plays s_B, the outcome is (s_A, s_B). Alice does not know Bob's strategy when choosing, so a belief μ_A over S_B turns each Alice strategy into a lottery over outcomes.

Under the VNM assumptions, we do not need to list Alice's preference over every possible lottery separately. We can summarize her preferences by a payoff function

$$u_A : S_A \times S_B \to \mathbb{R},$$

and similarly for Bob. This is the mathematical justification for putting numbers in payoff matrices.

Example 2.1. Let $S_A = \{T, B\}$ and $S_B = \{L, R\}$. If Alice believes Bob plays L with probability p and R with probability $1 - p$, then T gives the lottery

$$(T, L) \text{ with probability } p, \qquad (T, R) \text{ with probability } 1 - p,$$

while B gives the corresponding lottery over (B, L) and (B, R). The belief p is not fixed by the model; equilibrium analysis will later determine which beliefs and actions are consistent.

Example 2.2. Suppose Alice prefers T to B exactly when $p > 1/4$, is indifferent at $p = 1/4$, and prefers B when $p < 1/4$. Suppose also that she is indifferent between (B, L) and (B, R). A VNM representation is

$$u_A(T, L) = 3, \quad u_A(T, R) = -1, \quad u_A(B, L) = 0, \quad u_A(B, R) = 0.$$

Indeed, $U_A(T) = 3p - (1 - p) = 4p - 1$, while $U_A(B) = 0$. The sign of $4p - 1$ changes exactly at $p = 1/4$.

Alice	*L*	*R*
T	3	−1
B	0	0

Alice's VNM payoff representation in Example 2.2

2.4 Attitudes toward risk

Now specialize prizes to money and write $u : \mathbb{R} \to \mathbb{R}$. A fair gamble has expected monetary value 0. A risk-neutral decision maker is indifferent between accepting and rejecting every fair gamble, which happens exactly when u is linear. A strictly risk-averse decision maker rejects every nontrivial fair gamble, which corresponds to strictly concave u. Risk seeking corresponds to convex u.

> **Definition.** For monetary lotteries, a VNM utility function is:
>
> - risk-neutral if u is linear;
> - risk-averse if u is concave;
> - risk-seeking if u is convex.

For a concave utility function, the utility of expected wealth lies above the expected utility of risky wealth:

$$u(pW_1 + (1-p)W_2) \geq pu(W_1) + (1-p)u(W_2).$$

The vertical gap is the utility cost of risk. In money terms, it becomes the most the person would pay to replace the risky gamble with a sure amount.

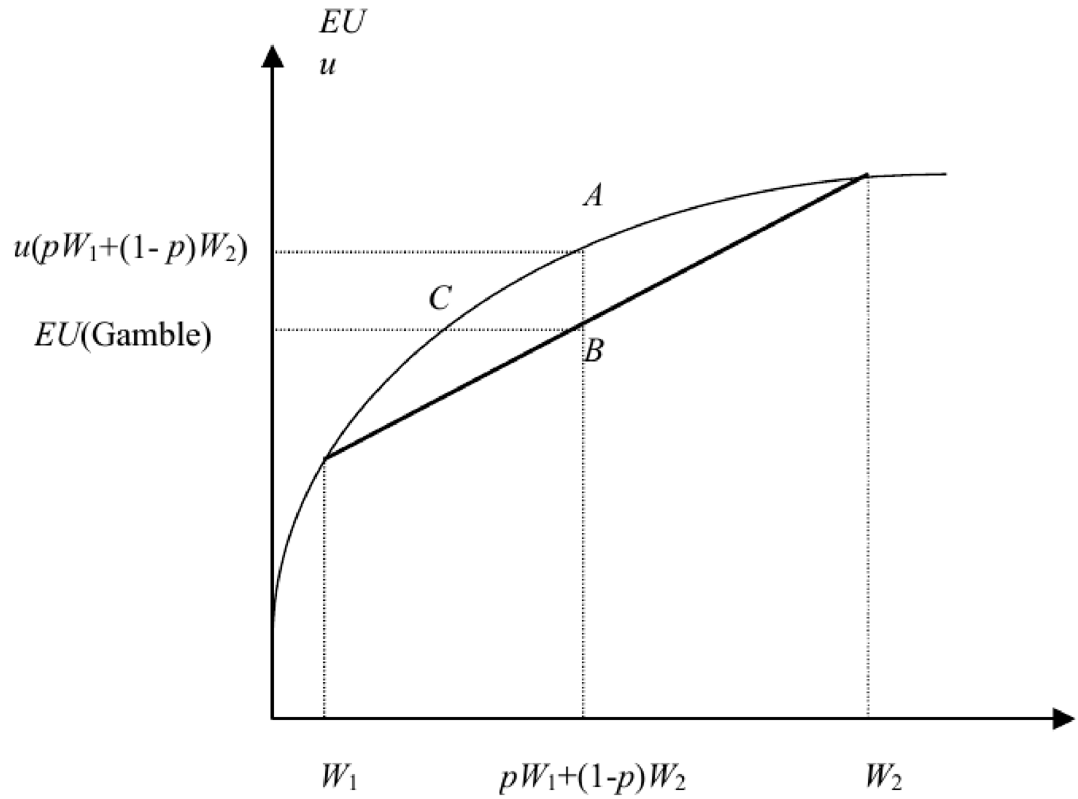


Figure 2.5:

CONCAVITY MAKES THE SURE EXPECTED VALUE MORE ATTRACTIVE THAN THE GAMBLE.

Risk sharing

Let $u(x) = \sqrt{x}$. A person owns an asset that pays 100 with probability $1/2$ and 0 with probability $1/2$. The expected utility is

$$\frac{1}{2}\sqrt{100} + \frac{1}{2}\sqrt{0} = 5.$$

If two identical people pool independent copies of this asset and each owns half the pool, each person's share pays 100 with probability $1/4$, 50 with probability $1/2$, and 0 with probability $1/4$. Expected utility becomes

$$\frac{1}{4}\sqrt{100} + \frac{1}{2}\sqrt{50} + \frac{1}{4}\sqrt{0} \approx 6.0355.$$

Risk sharing raises expected utility because the pool smooths each person's payoff.

Insurance

Keep the same risk-averse agent and add a risk-neutral insurer. Full insurance pays 100 exactly when the asset pays 0. If the premium is P, the insured person's wealth is $100 - P$ for sure. He buys insurance when

$$\sqrt{100 - P} \geq \frac{1}{2}\sqrt{100} + \frac{1}{2}\sqrt{0} = 5,$$

so $P \leq 75$. The risk-neutral insurer receives P for sure and expects to pay 50, so it is willing to sell when $P \geq 50$. Any premium in $(50, 75)$ benefits both sides.

Risk sharing and insurance

Move the high payoff. The asset pays the high amount with probability one half and zero otherwise; utility is $u(x) = \sqrt{x}$.

High payoff

$100

Standalone expected utility: **5**
Expected utility after pooling two independent assets: **6.04**
Mutually acceptable full-insurance premiums: $50to75$

Exercise 2.1. *Suppose two identical risk-averse agents can either buy insurance at a common premium or first form the mutual fund above. What range of premiums can make every party willing to participate?*

2.5 Takeaways

The chapter's modeling lesson is narrow but central. Payoff numbers in a game are VNM utility numbers. They summarize how a player evaluates lotteries over outcomes, and they are unique only up to positive affine transformations. As a result, replacing payoffs by an arbitrary increasing transformation preserves rankings of sure outcomes but can change preferences under uncertainty, and therefore can change the game.

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Chapter 3: Representation of Games

Based on Chapter 2 of 14.12 Economic Applications of Game Theory (MIT OpenCourseWare, Fall 2025).